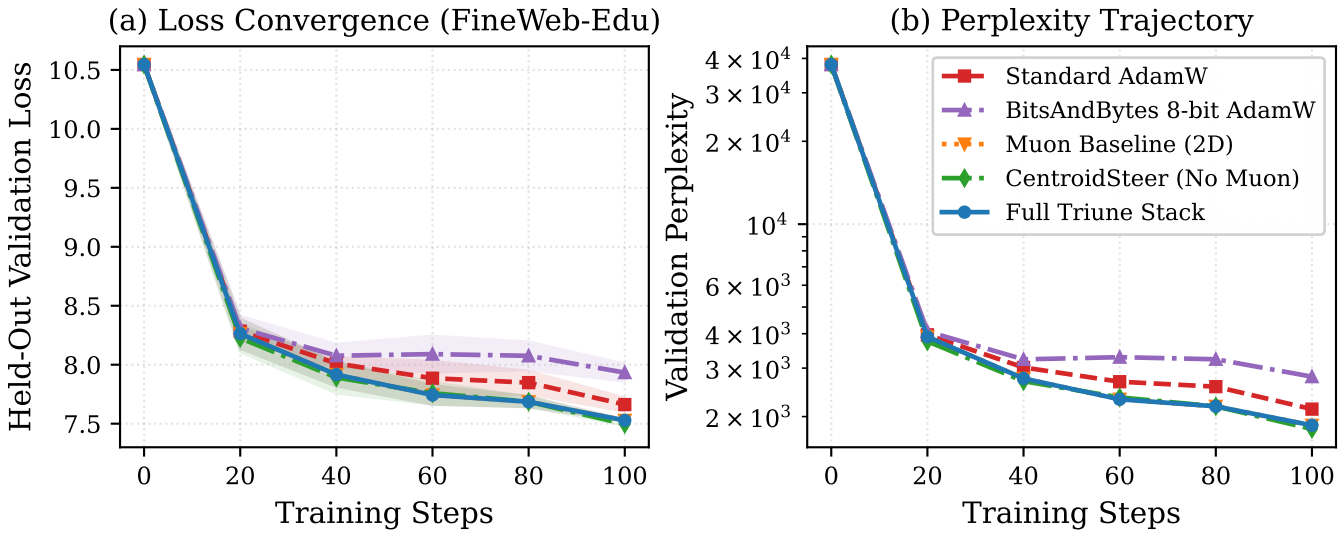
(a) Loss Convergence (FineWeb-Edu)
Held-Out Validation Loss
10.5
10.0
9.5
9.0
8.5
8.0
7.5
0
20
40
60
80
100
Training Steps
(b) Perplexity Trajectory
Validation Perplexity
4×10^4
3×10^4
2×10^4
10^4
6×10^3
4×10^3
3×10^3
2×10^3
Standard AdamW
BitsAndBytes 8-bit AdamW
Muon Baseline (2D)
CentroidSteer (No Muon)
Full Triune Stack
0
20
40
60
80
100
Training Steps